Laboratory Practical Work #1

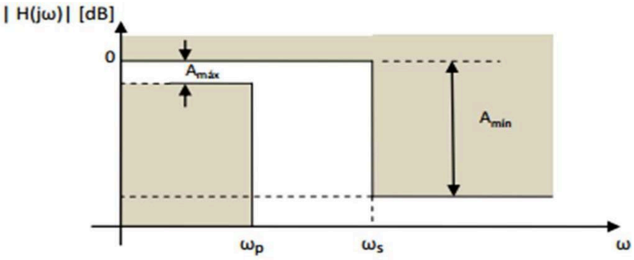
Active Analog Filtering

Goals:

- Identify the main characteristics of approximation functions
- Design active filters that meet the specified requirements.
- Handle circuit simulation and numerical tools.
- Implement the design and take measurements on it.

1. Graph the template with the design requirements. Indicate the assumed values. Graph the approximation function fitted to the template limits.
2. Obtain the filter order analytically, verify using Matlab, Python, etc.
3. Obtain the expressions that represent the group modulus, phase, and delay for the designed structure.
4. Draw the pole-zero diagram.
5. Obtain the frequency response (magnitude and phase) and the associated group delay using Matlab.
6. Obtain the value of the components for the required topologies.
7. Simulate the design using Workbench, Tina, or LTSpice and verify that it meets the requirements.
   Perform the simulation with ideal components and then replace them with commercial values and the A.O. to be used. Identify the differences. Justify your answer.
8. Plot the magnitude and phase graphs as a function of frequency.
9. Indicate the "setup" used in the measurement (type and model of instrument used, wiring diagram or photographs).
10. Conclusions: These should include the problems encountered throughout the process (design, simulation, assembly and measurement).

Design considerations



- Amáx = atenuación máxima permitida en la banda de paso, en [dB] (para Matlab "Rp")
- Amín = atenuación mínima requerida en la banda de atenuación, en [dB] (para Matlab "Rs")
- ωp = frecuencia límite de la banda de paso, en [r/s]
- ωs = frecuencia límite de la banda de atenuación , en [r/s]

Design specifications, simulation, assembly, and measurement

The design and manufacture of second-order structures must be done with structures**Ackerberg- Mossberg.**

Group	**Filter type**	**Approximation function**	**Requirements**
1	Bandstand	Equi-ripple	Ripple 3dB in the passband BW: 12kHz – 36kHz Minimum signal strength of 20dB for f < 4.8kHz. Minimum signal strength of 18dB for f > 72kHz.
2	High-passes	Equi-ripple	Passband: 3dB from 43.75kHz Stopband: 35dB from 12.5kHz
3	Eliminat e band	Maximu m Flatness	Passband: f<3.6kHz and f>9.1kHz Maximum At.: 1.5dB Rejection band: 5.45kHz<f<5.9kHz Minimum At.: 38dB
4	Underpasses	Bessel	Delay: 100us Maximum deviation: 10% at 20krad/s. Maximum gain: 1dB at 10krad/s
5	High-passes	Equi-ripple	Passband: 3dB from 35kHz Stopband: 35dB from 10kHz

6	Underpasses	Maximum Flatness	At. Máxima: 2dB a 10kHz Zero transmission at 45kHz and at infinity

Clarifications and recommendations

- During the design process, work in a standardized manner until the end to expedite calculations. Then, denormalize the values for standard resistor and capacitor values.
- If no restrictions have been given on some characteristic values, they must be assumed in the design process.
- The project is a group submission and the submission will be made in electronic format with the necessary files attached to reproduce the results obtained (matlab/python scripts, Tina/LTSpice files, etc.).
- You must implement the requested circuit on a board made by the group. It must not be on a breadboard or experimental board.
- Add measurement terminals to all components you deem necessary. You can do this with pinheads (like jumpers) to connect the oscilloscope.
- Use sockets for the integrated circuits.
- Add measurement and connection terminals for ground and power.
- Use an approximately 10uF electrolytic capacitor and another 100nF capacitor on your board's power supplies to reduce high-frequency noise that may be coming from the power supply.
- Leave terminal blocks or pins with enough space to connect alligator clips for the function generator and power supplies. It is preferable to use terminal blocks for the connection.
- Use presets for the resistors controlled by K and Q.